

The QA Process: A beginner's guide to the main stages, steps and tools of Quality Assurance | by Careerist | Medium

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The QA Process: A beginner's guide to the main stages, steps and tools of Quality Assurance When you're thinking about becoming a tester, and you're reading about the topic, you can quickly become overwhelmed with all the concepts and terms associated with QA. There are so many terms in the QA process that don't mean anything to you, like 'black box', 'white box', 'unit tests', 'regression tests', 'manual' and 'automated' testing, 'BrowserStack', and 'Jira'... It all sounds so complicated! Plus, you probably think that it'll take you years to learn all of this stuff and to become a successful QA engineer! Well, we won't lie to you and say that becoming a QA engineer is simple and quick, because becoming a tester is going to require effort. But believe us when we say, it is not that hard to actually become a QA engineer. With adequate guidance and training you can learn pretty fast. You just need to know where to start. This article will help you to understand the QA process, including, the testing steps, and what tools are generally used by engineers. After reading this article you will know: What is the difference between quality assurance and quality control? What are the main stages in the QA process? What is the difference between manual and automated testing? What do the different 'test types' mean? Which tools help to optimize the entire QA process? What is the QA process and how is it different from quality control? QA, or quality assurance, in software development is a process that checks and prevents any issues occurring across the whole project. This also includes project management, as various individuals need to be managed when performing QA tests. And finally, QA also guarantees that a company delivers a product or software to market that has the best possible user experience. Quality control (QC), on the other hand, focuses on testing the final product and identifying issues and eliminating them, before the product goes on the market. Having a functioning QA process ensures that all of the team members, who are working on a particular project, work more effectively together and in doing so the development time will shorten, and thus costs will be lower. Of course, the QA testing process varies slightly in different companies. But in general, its stages are the same. Now you know more about the QA process let's take a look at the main stages within the QA process. What are the main stages in the QA process? Review of requirements Nowadays, quality assurance starts very early in the software development lifecycle – right in the 'requirement analysis' stage. QA professionals must review the requirements of the product before starting work, so that they can be clear on what needs to be done. Checking the requirements also means that the testing will be consistent, complete, executable, and that the project is actually testable. Planning phase After testers have understood the requirements, they can start developing a test strategy and begin the test planning phase. At this stage, they define the scope of the work, the budget, they decide which approach to take at each stage of the software's development, they'll also decide what types of testing will be needed, and finally, which tools will be used during the actual testing. Developing test cases With a plan in hand, it is time to design test cases, create checklists, prepare the environment for test execution, and create automated testing scenarios. Test execution At this point, everything is prepared for finding bugs and issues. The QA team starts to execute different types of tests, and testers start to report any and all of the discovered bugs. Verifying fixes Once the development team fixes an issue, QA professionals retest the functionality of the product, and perform regression testing to ensure that the software still works correctly after all of the changes have been made. Test cycle closure Once all of the fixes have been changed, and all planned tests have been executed, it is time to prepare the test closure document. This document is a summary of all the tests performed

during the software development life cycle. What tests are used in the QA process, and what approaches are considered? Now that you understand what the QA process looks like, let's now talk about all the different test types that can be used in software testing.

Yes, there are a lot of them, but don't worry, you'll soon be able to navigate between them.

Functional and non-functional tests

There are two main tests categories, these are, functional and non-functional tests. In functional testing, QA engineers check to see whether the product behaves as required. In other words, they verify if the actual output is matching the expected result. In non-functional testing, the QA engineer checks how the product performs in different conditions. They will change the environments to see what happens, for example, load, security, stress, and usability tests.

Knowledge of the source code

If testers know the source code before testing begins, then QA engineers use the 'white box' testing type. On the other hand, if QA engineers do not know the source code they are dealing with, the 'black box' testing type is used. This is when testers only evaluate input and output in an application. On a different note, 'grey box' testing is a combination of both of these approaches, and testers use this type of testing when they have limited information about the internal structure of the system.

A phase of software development

This is when tests are divided up into; unit testing, module testing, integration testing, system testing, and user acceptance testing, based on where developers are in the software development cycle.

Here are a couple more testing types that you may come across:

- Smoke tests are designed to check the basic functionality of an application. They are quick to perform, and with their help, testers can ensure that the main features of the system are working correctly.
- Regression tests help to check if an application is working as it is supposed to be working after some changes, such as bug fixes, are done.
- Load tests are needed to determine an application's behavior both under average and peak load conditions.
- Cross-browser/cross-platform testing helps to analyze the application's behavior in different browser/platform environments.

Of course, these are not all of the testing types used in the QA test process. But knowing the basic categories will help you to understand the topic better.

Approaches to executing tests

Some tests are performed by humans and this is known as manual testing. In manual testing, testers actually execute test cases and generate test reports themselves. On the other hand, you have tests that are performed by computers, this is more commonly known as automation. Test automation engineers are responsible for creating and writing test codes that simulate user experiences, that can be repeated over and over again, to check the software in question for bugs. Both of these approaches have pros and cons, and when used together they yield good results. Here are some things about manual testing to consider.

Manual testing

Manual testing is more suited for small test projects, and manual tests typically pick up issues that would be missed in an automated test. And finally, manual testing doesn't require advanced knowledge of scripting languages and is reasonably easy to learn. On the flip side, manual testing is time-consuming if testers are working on a big project, and you can never exclude the possibility of human error. Similarly, automation has many positives, it can be used repeatedly, time and time again. Plus, it's good for large projects. But, on the downside, it doesn't quite have the human touch and can miss errors that are beyond the scope of its code. QA professionals have to decide between manual and automated testing for each project.

Tools

Now that we know what the QA process is and what types of tests and approaches exist, let's talk about what tools testers use when they're completing their work. A testing tool is anything that supports test activities. There are plenty of tools on the market, and nobody expects a beginner to know them all. But it's a good idea to be familiar with the most popular tools, like 'Selenium', 'Jira', and 'BrowserStack'.

- 'Selenium' is the most popular testing tool. It does not need advanced scripting knowledge and is user friendly for beginners.
- 'Jira' is a bug and issue tracking tool. It is also used for project management.
- 'BrowserStack' enables developers to test their applications across different browsers, devices, and operating systems.

Conclusion

No matter which approach or method a company uses, the final goal is always the same – to

deliver the best possible product to customers. Well established QA processes help to reduce costs, improve the overall quality of software, and bring about a better user experience. You can become part of this process right now by signing up to one of our online courses. Our short courses are designed to help people just like you to overcome their fears and start a new career as a QA professional. With our experienced tutors guiding you along the way, you'll have learnt the basics of QA in just a few weeks – then you'll be on course to get yourself a high-paying salary. Check out our website for more information, and to learn about our courses and curriculums right now.